AI Study Assistant (RAG-Powered Question-Answering System)

1. Introduction

The AI Study Assistant is designed to help students understand academic content by answering questions and generating summaries from user uploaded documents. The system follows a hybrid architecture that combines a fine tuned T5 model with a powerful pretrained Qwen model. The aim is to use the custom trained T5 model as the primary generator since it is tailored to question answering tasks, while Qwen acts as a secondary reasoning model when more complex interpretation is required.

The system supports both question answering and document summarization through an intuitive graphical interface. It processes lecture notes, research papers, PDF files, and text based content, enabling students to study more efficiently.

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2. Objectives

1. To build a practical AI based study tool capable of answering academic questions.
2. To train a custom transformer based model (T5) for question answering.
3. To incorporate a fallback reasoning model (Qwen 2.5 1.5B Instruct) for improved depth.
4. To implement an efficient retrieval pipeline for extracting relevant document sections.
5. To provide a simple user interface for question answering and summarization.

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3. Dataset

The system was trained using two public question answering datasets:

1. **SQuAD v2.0**
   Contains questions that may or may not have answers within the provided context.
   Format: paragraph, question, and correct answer.
2. **SciQ Dataset**
   Consists of science related questions with short factual answers.

Both datasets were preprocessed into a unified instruction style format suitable for T5 fine tuning. No additional datasets were used.

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4. Preprocessing Steps

1. **Question and Context Pairing**
   Input format created as:
   qa: question: <question> context: <context>
2. **Answer Normalization**
   Answer text converted to plain strings and cleaned.
3. **Tokenization**
   Input and target sequences tokenized using T5 tokenizer with fixed maximum lengths.
4. **Label Masking**
   Padding tokens replaced by -100 to prevent loss calculation on padded regions.
5. **Shuffling and Splitting**
   Training and validation sets prepared for stable fine tuning.

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5. Model Architecture

The system follows a hybrid architecture composed of:

5.1 Fine Tuned T5 Model

A transformer based sequence to sequence model used as the primary QA generator. It was trained on SQuAD and SciQ to specialize in short answer extraction and concise explanation.

5.2 Qwen2.5 1.5B Instruct Model

Used as a fallback reasoning model. It handles cases where the T5 output may be vague or incomplete. Qwen refines answers using richer instruction understanding.

5.3 Retrieval and Embedding

The system uses:

- **HuggingFace Embeddings** to convert document chunks into embeddings.
- **FAISS Similarity Search** to find the most relevant chunks.
- **Cosine Similarity** for ranking.

This retrieval enhanced generation (RAG like behavior) ensures that answers are grounded in the uploaded document.

5.4 Decoding

Beam search decoding used for improved output quality in T5 and Qwen.

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6. Algorithms Used

To keep the system efficient and grounded, the following core algorithms were used:

1. **Transformer Sequence to Sequence Learning**
2. **Multi Head Attention for contextual alignment**
3. **Beam Search for controlled generation**
4. **Similarity Search using FAISS + Cosine Distance**
5. **Two Stage Hybrid Answering Logic**
    - Stage 1: T5 produces answer
    - Stage 2: Qwen refines if needed

These algorithms directly contribute to accurate and context aware responses.

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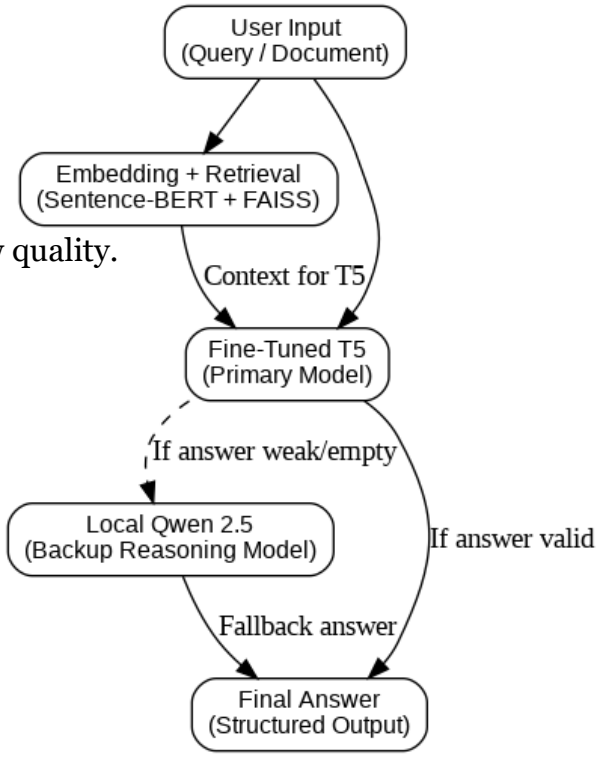
7. Hybrid Model Workflow

1. User uploads a PDF or document.
2. Document text is chunked and embedded.
3. User enters a question.
4. Relevant chunks retrieved through FAISS.
5. T5 generates the first answer.
6. System checks if the answer is empty, generic, or low quality.
7. If required, Qwen generates a refined answer.
8. Final output returned to the user.



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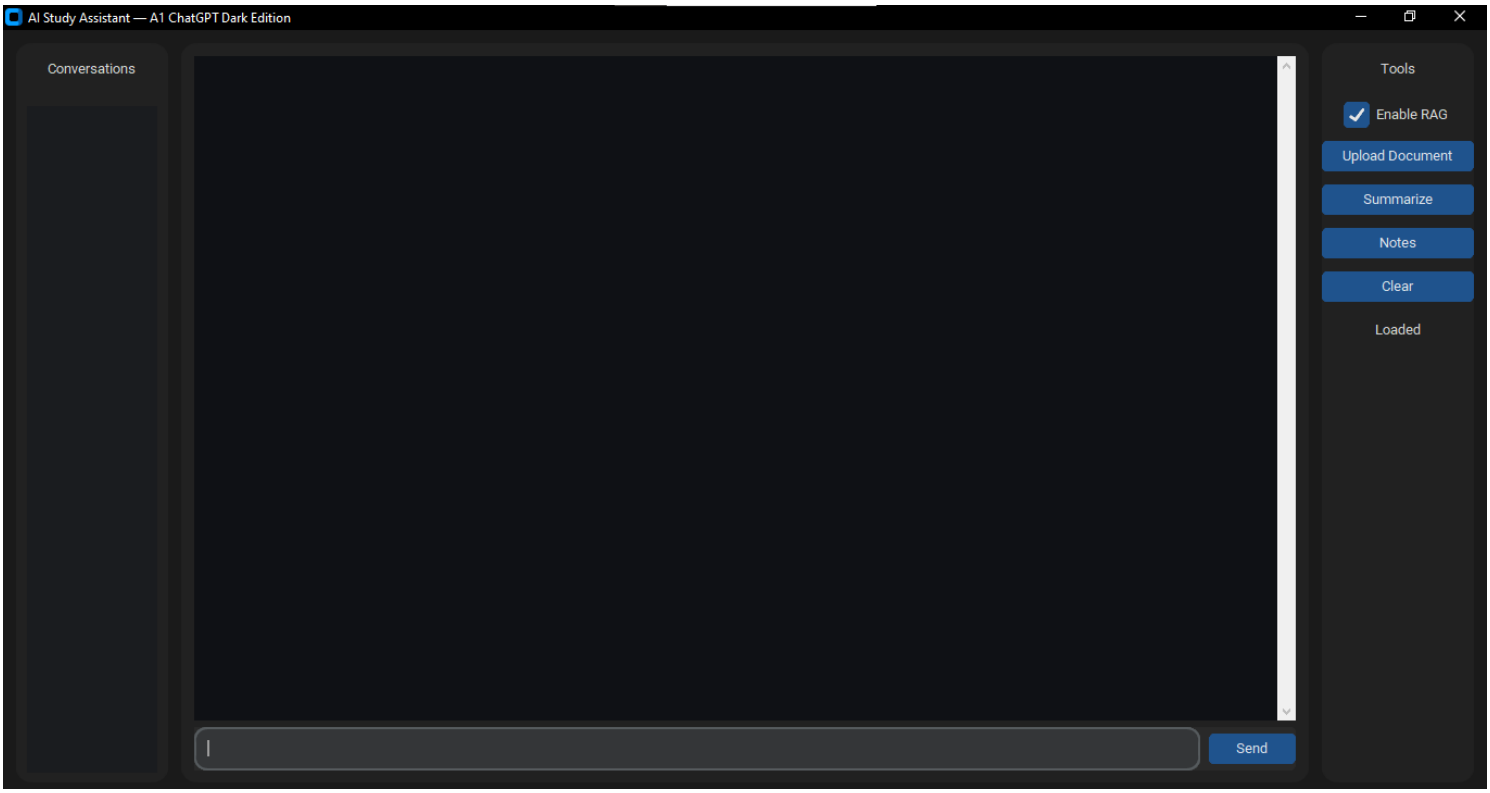
8. Graphical User Interface

The interface allows:

- PDF upload
- Question answering
- Document summarization
- Viewing generated results



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9. Model Evaluation

Since the system is a **generative question answering model**, traditional metrics like accuracy or confusion matrix are not meaningful. These metrics work only for fixed classification outputs, not for natural language generation.

Why Accuracy and Confusion Matrix Cannot Be Used

- Answers are full sentences, not discrete labels.
- Two answers may be worded differently but mean the same.
- Exact match accuracy punishes small wording differences.
- Confusion matrices require fixed classes, which generative models do not have.

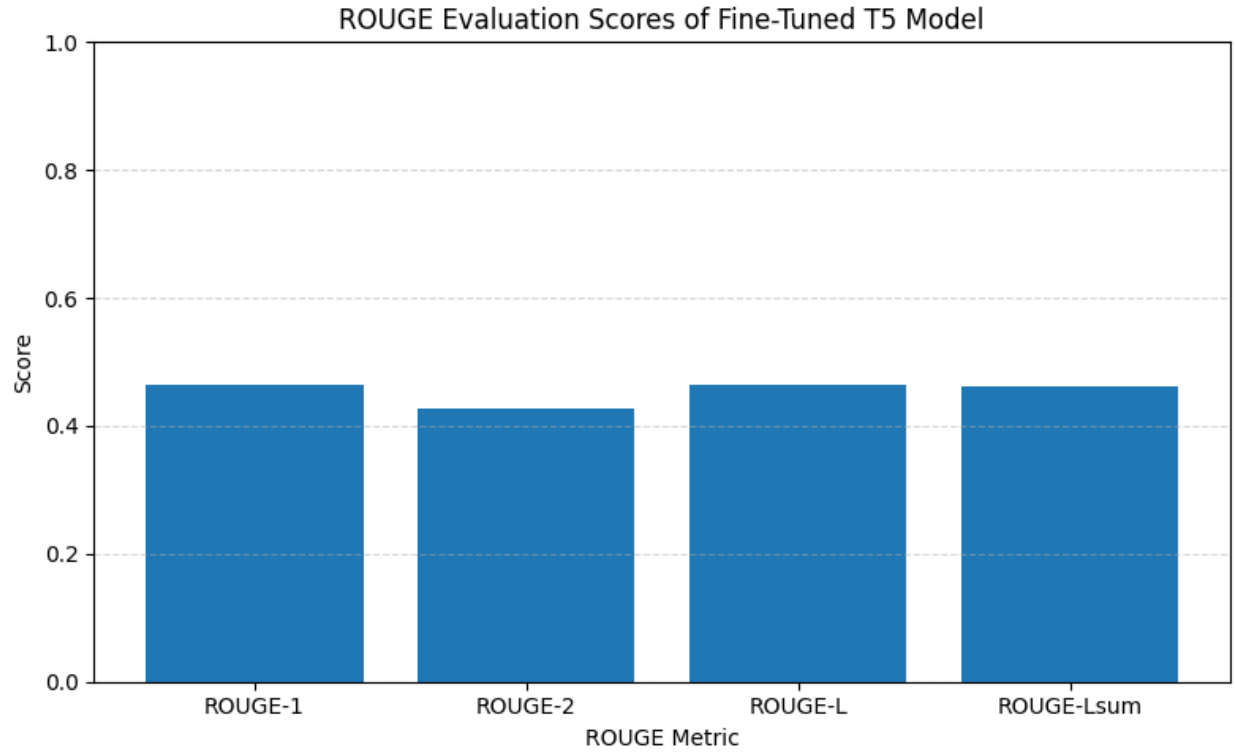
Correct Evaluation Metrics Used

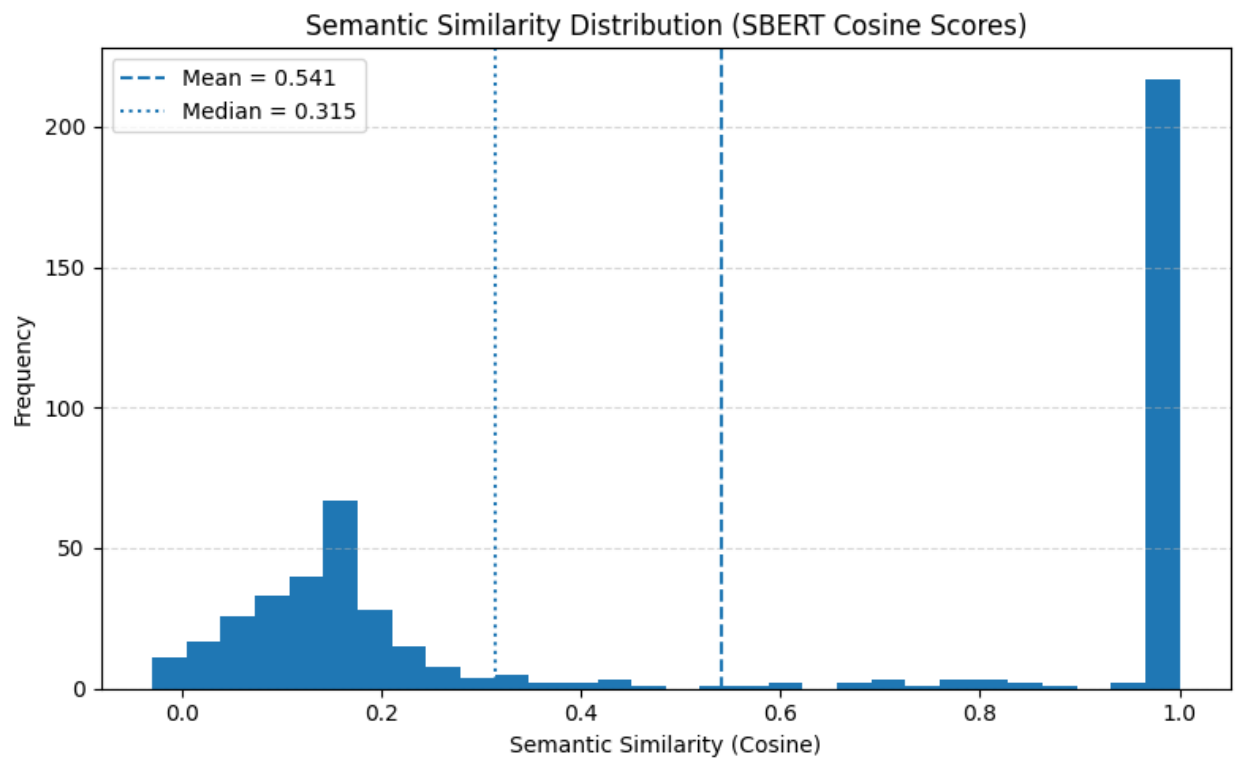
1. **ROUGE Scores** for lexical overlap.
2. **Semantic Similarity (Sentence Transformer)** for meaning based similarity.

Evaluation Results

- ROUGE 1: 0.464
- ROUGE 2: 0.428
- ROUGE L: 0.463
- Average Semantic Similarity: 0.54

These scores indicate that the model generates answers semantically close to ground truth and performs reasonably for a lightweight fine tuned T5 model.





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10. Why Semantic Metrics Are More Suitable

Generative models produce free form answers, so evaluating meaning is more important than matching exact words.

- ROUGE identifies phrase overlap.

- Semantic similarity captures meaning, even if wording differs.
- These metrics are widely used in summarization and QA tasks.

This makes them a better fit for realistic performance evaluation of transformer models like T5.

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11. System Strengths

- Works with multiple document types.
- Hybrid design improves answer quality.
- Lightweight fine tuned model reduces computation.
- Strong fallback mechanism through Qwen.
- Simple and accessible GUI.

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12. Limitations

- Qwen inference is slow on CPU based systems.
- T5 may struggle on very long contexts.
- Performance depends on the clarity of the uploaded document.
- No GPU acceleration in the deployed environment.

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13. Conclusion

The AI Study Assistant successfully combines a custom fine tuned T5 model with a powerful Qwen model to create an efficient study companion tool. By integrating retrieval based context selection, generative QA, and semantic evaluation, the system provides reliable academic assistance while remaining computationally lightweight.

Its hybrid structure balances specialization and general reasoning ability, making it suitable for real world educational use.